

Question 1 : Write short answer on any of the follow :

(1+1+2 m)

(a) When we use the *slack* variable?

(b) When we use the *surplus* variable?

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(c) **Fill in the blanks:** In LPP with n variables and m constraints. If the $m \times m$ square system of standard form has a unique solution, this solution is called a **basic solution**. Further if it is feasible, it is called a **Basic Feasible**.

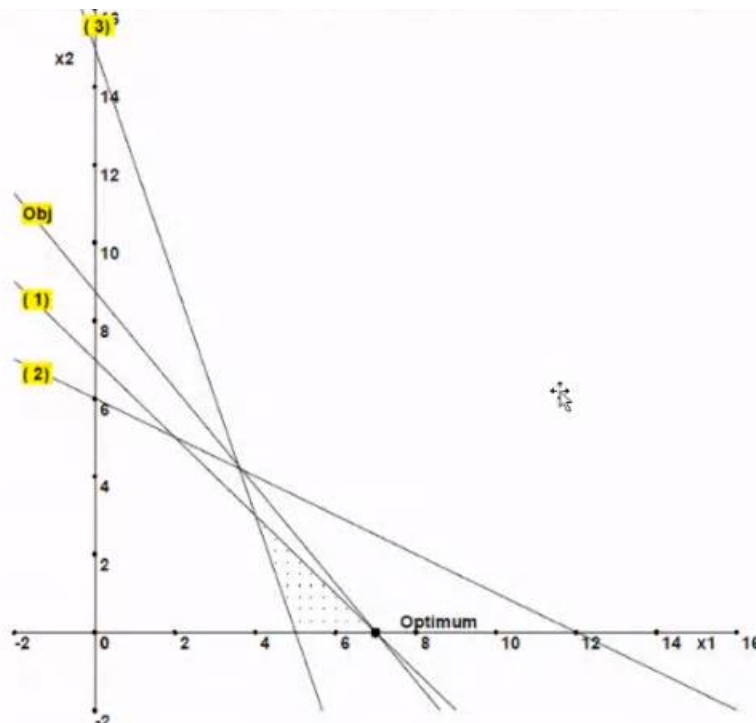
Question 2 : Solve the problem Graphically .

(7 m)

$$\text{Maximize } Z = 5x_1 + 4x_2$$

$$\begin{aligned} \text{S. T : } & 2x_1 + 2x_2 \leq 14 \\ & x_1 + 2x_2 \leq 12 \\ & 3x_1 + x_2 \geq 15 \\ & x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

Objective Value = 35.00 at $x_1 = 7.00$ and $x_2 = 0.00$



Question 3 : Given the following problem**(5 m)**

$$\text{Maximize } Z = 2x_1 + 3x_2$$

Subject to:

$$\begin{aligned} x_1 + 2x_2 &\leq 15 \\ -2x_1 + 4x_2 &\geq 20 \\ x_2 &= 5 \\ x_1 \geq 0, x_2 &\text{ Unrestricted} \end{aligned}$$

Write the Problem in standard form.

$$\text{Max } Z = 2x_1 + 3x_{2+} - 3x_{2-} + 0s_1 + 0s_2 = 0$$

s.t:

$$\begin{aligned} x_1 + 2x_{2+} - 2x_{2-} + s_1 &= 15 \\ -2x_1 + 4x_{2+} - 4x_{2-} - s_2 &= 20 \\ 2x_{2+} - 2x_{2-} &= 5 \\ x_1, x_{2+}, x_{2-}, s_1, s_2 &\geq 0 \end{aligned}$$

Question 3 : Given the following problem**(2+7 m)**

$$\text{Maximize } Z = 20x_1 + 10x_2$$

Subject to

$$\begin{aligned} x_1 - x_2 &\leq 1 \\ 3x_1 + x_2 &\leq 7 \\ x_1 \geq 0, x_2 &\geq 0 \end{aligned}$$

- a) Write the problem in standard form.
b) Solve the problem using the algebraic method.

$$\begin{aligned} Z - 20x_1 - 10x_2 + 0s_1 + 0s_2 &= 0 \\ x_1 - x_2 + s_1 &= 1 \\ 3x_1 + x_2 + s_2 &= 7 \\ x_1, x_2, s_1, s_2 &\geq 0 \end{aligned}$$

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Nonbasic (zero) variables	Basic variables	Basic solution	Feasible?	Objective value, z
(X1,X2)	(S1,S2)	(0,0,1,7)	Yes	0
(X1,S1)	(X2,S2)	(0,-1,0,8)	Yes	-10
(X1,S2)	(X2,S1)	(0,7,8,0)	Yes	70 Optimal
(X2,S1)	(X1,S2)	(1,0,0,4)	Yes	20
(X2,S2)	(X1,S1)	(7/3,0,-4/3,0)	No	-
(S1,S2)	(X1,X2)	(2,1,0,0)	Yes	50

- ☐ The optimal solution to the given problem as $x_1 = 0, x_2 = 7$
☐ Optimal $z = \text{Maximum} = 70$